

STANDALONE INSTRUMENT + VST3 / AU PLUG-INS



NSeqArpKeys

PATTERN INSTRUMENT

User manual for the pattern-driven keyboard instrument.

Assign a sequence and a Forte pitch-class set to any MIDI key. Hold the key to play the pattern, or latch it while you edit. Shape note lengths with Gate and Fixed Steps, then save the complete setup as a preset.

1. Get started

You can use NSeqArpKeys on its own or load a plug-in in a compatible DAW.

Install

Standalone

Extract your OS archive and run

`NSeqArpKeys.exe` on Windows, `NSeqArpKeys` on Linux, or `NSeqArpKeys.app` on macOS.

Use **Options** to select an audio device if needed.

Plug-in

Copy the complete `NSeqArpKeys.vst3` bundle to your VST3 folder and rescan in your DAW.

The macOS archive also includes `NSeqArpKeys.component` for Audio Units.

Keep plug-in bundles intact. VST3 folders: Windows `C:\Program Files\Common Files\VST3`, Linux `~/vst3`, macOS `~/Library/Audio/Plug-Ins/VST3`. For Audio Units on macOS, use `~/Library/Audio/Plug-Ins/Components`.

Make your first pattern

- 1 Open **Browse Presets**, choose **Single Note Pulse**, and click **Load**. This assigns a playable example to C4 (MIDI 60).
- 2 Hold C4 on the on-screen keyboard or send MIDI note 60 to the plug-in. The pattern plays while that key is held.
- 3 Release C4 to stop. To keep it running while editing, turn on **Latch (keep playing)**, trigger C4 once, and release it.
- 4 Click the same key again, or use **Stop Key** or **Stop All**, to stop a latched pattern.

A blank key needs both a nonzero Pattern step and a Forte Set before it produces notes. The Init preset is deliberately silent.

2. Work in the editor

The keyboard selects a trigger note. The controls below it edit the assignment for the currently selected key. Each of the 128 MIDI keys can have its own pattern, set, channel, octave, gate, and fixed step length.

Control	What it does
Preset name / < / >	Shows the loaded preset and steps through the preset library. An asterisk means the current setup has changed.
Browse Presets / Save	Opens the library. Save updates the current user preset, or opens the browser to save a new copy.
Keyboard	Triggers a pattern and selects the MIDI key whose assignment is shown.
Latch / Stop Key / Stop All	Keeps patterns running after release, stops the selected key, or stops every active key.
Meter Num / Meter Den	Global meter settings. Meter Den controls the duration of one pattern step at the current tempo.
Channel / Octave	Sets the selected key's MIDI output channel and pitch region.
Gate / Fixed Steps	Adds a gated fraction of the span through rests and a fixed duration in steps.
Pattern	Space-separated step values. Zero creates a rest and extends the previous sounding step's available gate span.
Find Set / Forte Set	Search by Forte ID, common name, or pitch classes, then choose a matching pitch-class set.

Change a running pattern

With Latch on, trigger a key and release it. Edit Pattern, Gate, Fixed Steps, Forte Set, Channel, or Octave while listening. The active pattern restarts with the new assignment on the next audio block.

Selected key matters. Stop Key applies to the key shown in the selected-key label. Stop All ends every currently playing pattern.

3. Patterns, pitch, and timing

Pattern values

Enter whole numbers separated by spaces. Each number is one step. Positive numbers are bit masks over ascending notes of the selected Forte set. Negative numbers select in the opposite direction from the Octave base. Zero starts no new note.

Example - four steps

1 2 3 0

1 plays the first available note, 2 the next, and 3 plays both together. 0 creates a rest after the chord.

Octave moves the base position used by these bit masks. Channel sets the outgoing MIDI channel from 1 to 16. Notes outside MIDI's 0-127 range are ignored.

Gate across zero steps

A sounding step owns the zero steps that follow it, including zero steps at the end of a looping pattern. Gate is a percentage of that entire span, from the note onset to the next sounding step.

Example - a three-step span

1 0 0 2

The first note has three steps available: its own step and two following zeros. At 50% Gate it sounds for 1.5 steps; at 100% it reaches the next sounding step. A zero step does not retrigger the note.

Gate ranges from 0% to 200%. Values above 100% can overlap later notes. Fixed Steps ranges from 0 to 16 in increments of 0.01. The two lengths add: $\text{note length} = \text{Fixed Steps} + \text{Gate} \times \text{sounding span}$. For example, a three-step span with Gate 50% and Fixed Steps 0.5 lasts 2 steps. Releasing the trigger key stops the pattern immediately when Latch is off.

Tempo and meter

Pattern steps follow the DAW tempo when a host supplies one. Otherwise the instrument uses 120 BPM. One step lasts $60 / (\text{BPM} \times \text{Meter Den})$ seconds: at 120 BPM and Meter Den 4, a step lasts 0.125 seconds. Meter Num is saved with the setup but does not currently change step duration or pattern length.

4. Forte sets

A Forte set supplies the pitch classes from which Pattern values choose notes. Each menu match shows its Forte ID, any common name, and its pitch classes in brackets.

1. Select the MIDI key you want to configure.
2. Type part of an ID, a common name such as `diatonic`, or pitch classes in **Find Set**.
3. Open **Forte Set** and choose a match. The selected set is shown underneath.

Example

7-35.11 - diatonic scale [0 2 4 5 7 9 11] contains seven pitch classes. With this set, 1 2 4 8 walks upward through its first four scale positions.

Choose **(No set)** to clear the selected key's set. A pattern without a set is silent. Changing the search text only filters the menu; choose a result to change the assignment.

Live performance

Normally, a pattern runs only while its trigger note is held. Latch changes note release behavior: the first trigger keeps the pattern running, and another trigger of the same key switches it off. Turning Latch off stops patterns whose trigger keys are no longer held.

The plug-in emits generated MIDI notes and also has an internal preview sound. In a DAW, generated MIDI routing depends on the host. In standalone mode, check **Options** for the audio output device if you cannot hear the preview.

5. Presets

A preset captures the meter, latch setting, selected key, and all 128 key assignments. The library has six factory examples and can hold your own presets.

Browse and load

Open **Browse Presets**. Search names, categories, tags, and descriptions; narrow results with the Category menu or **Favourites only**. Select a row to inspect its details, then click **Load**. Loading replaces the current setup and stops any active patterns.

Save and manage

Action	Result
Save New	Stores the current setup under the name, category, tags, and description shown on the right.
Save	Updates the currently loaded user preset. For a factory preset, opens the browser with a suggested copy name.
Update	Writes the current setup and edited details to the selected user preset. Factory presets cannot be updated.
Duplicate	Creates and loads a user copy of the selected preset.
Favourite / Unfavourite	Adds or removes the selected preset from the favourites filter.
Delete	After confirmation, removes a user preset from disk. Factory presets cannot be deleted.
Import / Export	Copies a <code>.nseqpreset</code> file into the library, or saves the selected library preset to a file. Import selects the new row; click Load to apply it.

Where presets live: in the OS application-data folder under `NSeqArpKeys/Presets` (for example, `%APPDATA%\NSeqArpKeys\Presets` on Windows). The standalone app and plug-ins share this folder on each OS.

6. If something is not working

What you notice	What to check
No notes when a key is held	Select the key, enter at least one nonzero Pattern value, and choose a Forte Set. Try the Single Note Pulse factory preset on C4.
The pattern stops when you release the key	That is the default behavior. Turn on Latch before triggering the key if you want it to continue during edits.
A pattern keeps playing	Click its trigger key again while Latch is on, use Stop Key for the selected key, or use Stop All.
No sound in standalone mode	Open Options and select an active audio output device. Confirm the selected key has a playable assignment.
VST3 does not appear in the DAW	Keep the outer <code>NSeqArpKeys.vst3</code> folder intact in the VST3 directory and rescan plug-ins in the DAW.
A preset does not appear	Check the browser's search, category, and Favourites only filters. User files must be valid <code>.nseqpreset</code> files in the preset folder.

Quick reference

To hear a pattern

Assign Pattern + Forte Set, then hold the trigger note. Use Latch to keep it running while you edit.

To share a setup

Save New, select the preset in the browser, then Export. Send the resulting `.nseqpreset` file.

NSeqArpKeys 1.1.1 | Windows, Linux, macOS | Standalone and VST3; Audio Units on macOS.